Lab 5: Data Flow Diagram

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Project: AI Chatbot for University Helpdesk

1. Level 0 DFD (Context Diagram)

The whole system is shown as a single process that interacts with three external entities: Student, Administrator, and the LLM API.

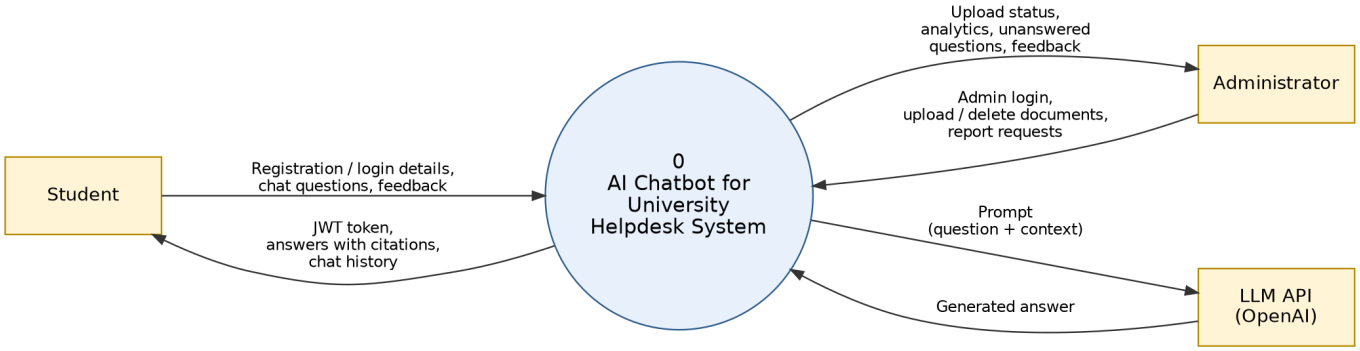


Figure 5.1: Level 0 DFD of the AI Chatbot for University Helpdesk

2. Level 1 DFD

The system is decomposed into five processes: Authenticate User, Process Chat Query (RAG), Manage Knowledge Base, Collect Feedback, and Generate Analytics. Data stores: D1 Users, D2 Conversations, D3 Document Metadata, D4 Vector Store (ChromaDB), D5 Feedback & Unanswered.

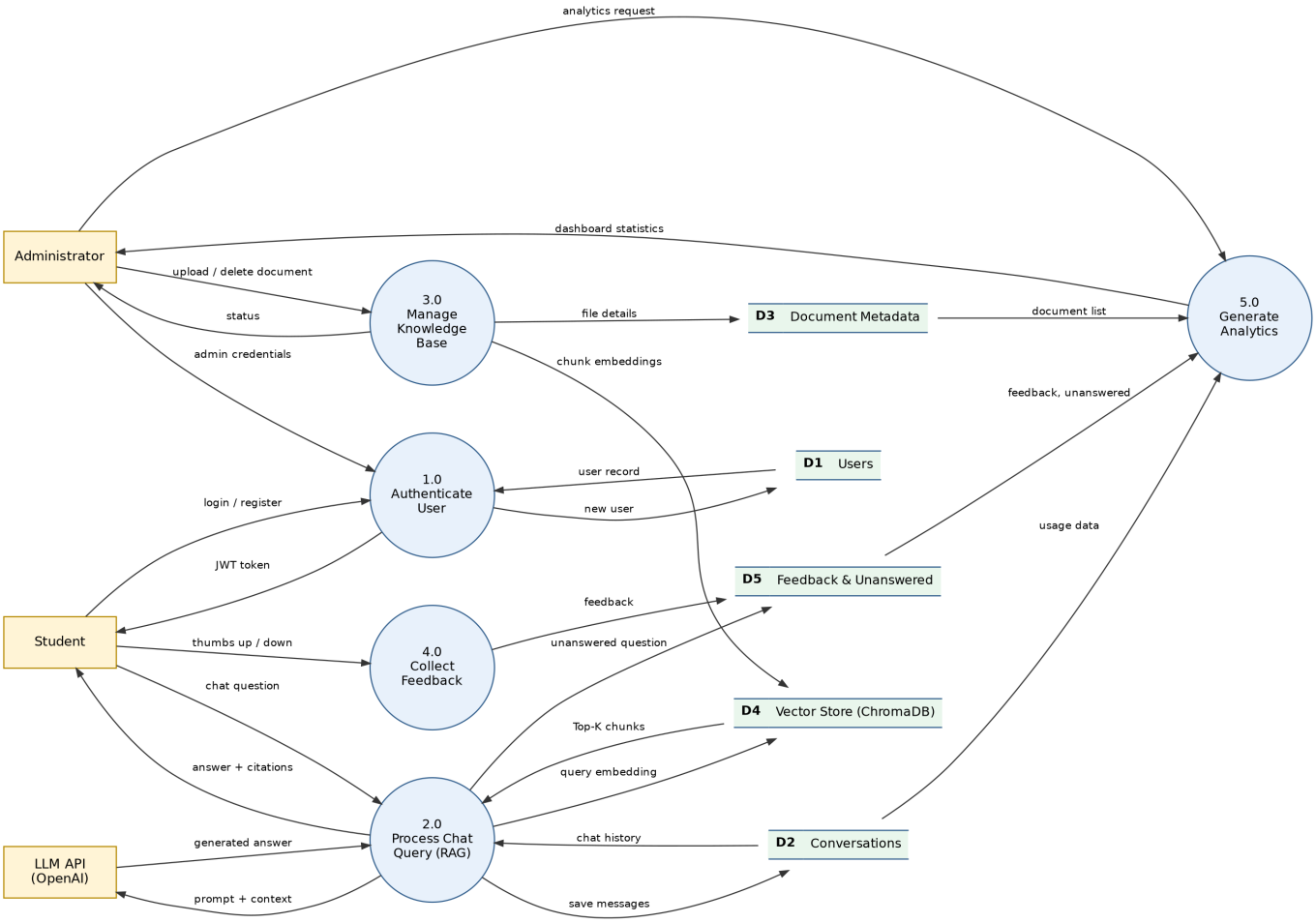


Figure 5.2: Level 1 DFD